

AI as an R Assistant - Prompt Card

Optional SEAC 2026 workshop extension

Use AI to adapt and troubleshoot **validated workshop code**, not to silently replace the statistical method.

1. Good uses

- change graph titles, labels and text size;
- change PNG dimensions/resolution;
- adapt the script to different column names;
- explain an R error in plain language;
- identify a missing package/object;
- explain what a line of R code does.

2. Safe prompt structure

Give:

1. **Context** - which workshop script/method you are using.
2. **Goal** - the exact change you want.
3. **Evidence** - relevant code and complete error/output.
4. **Constraint** - what must not change.

3. Customise a graph

I am using validated R code from a circular-statistics workshop.
The statistical calculation is correct.

Please change only:

- title to: [TITLE]
- x-axis label to: [LABEL]
- base text size to 14 pt
- output to a 300 dpi PNG, 180 mm wide

Do NOT change:

- the analysed variable;
- circular versus linear treatment;
- kernel;
- bandwidth;
- uncertainty treatment;
- normalisation;
- statistical test.

Return the complete corrected R code and mark the lines changed.

[PASTE CODE]

4. Fix an error

I am running this validated R workshop code in RStudio on Windows.

Code:

[PASTE RELEVANT CODE]

Complete error message:

[PASTE ERROR]

Please:

1. explain the error in non-technical language;
2. identify the most likely cause;
3. give the smallest correction;
4. return the corrected code block;
5. do not replace the statistical method.

5. Adapt column names

The workshop script expects:
[LIST EXPECTED COLUMNS]

My data contain:
[PASTE colnames(mydata)]

Tell me exactly which lines to change so the same analysis works.
Preserve all statistical calculations and parameter values.

6. Evidence to provide when debugging

Useful commands:

```
sessionInfo()  
getwd()  
colnames(dat)  
head(dat)
```

plus the exact error and the smallest relevant code block.

7. Never let AI silently change

- analysed variable;
- degrees versus radians;
- circular versus linear;
- directional versus axial;
- kernel;
- bandwidth/multiplier;
- uncertainty meaning;
- normalisation;
- significance level;
- sample filtering;
- missing-data handling.

8. Useful final sentence

If the requested change requires altering the statistical method, do not alter it silently. Explain why first and keep the original method unless I explicitly approve the methodological change.

9. Privacy

Use the workshop's synthetic datasets for practice. Before sending unpublished or restricted research data to an online AI service, follow your institution's rules and the applicable service/account settings.

10. Human verification

After accepting AI-generated R code:

1. run it;
2. check warnings/errors;
3. compare with the known-working version;
4. confirm that the method and parameters are unchanged;
5. save the final code with the research files.

Use AI to reduce coding friction, not to outsource methodological judgement.